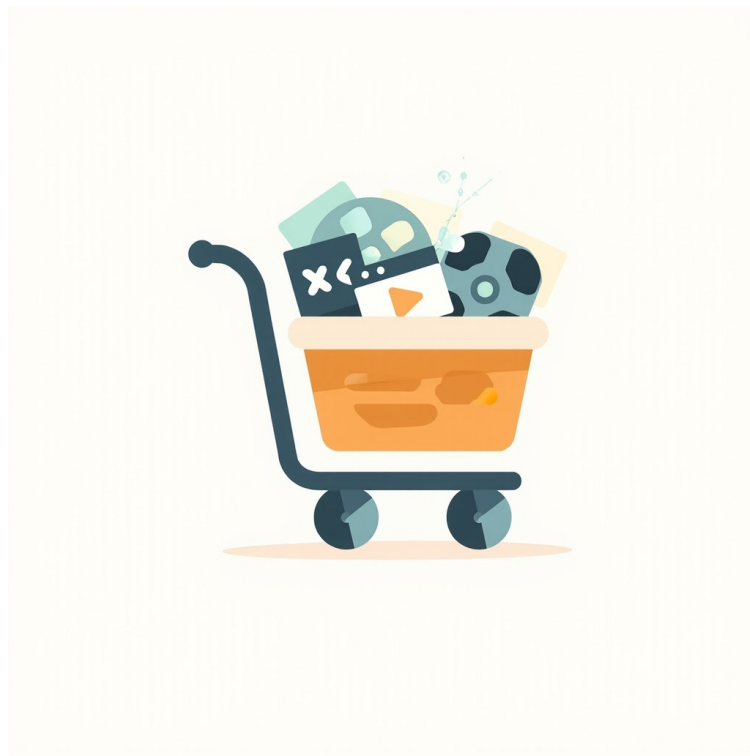


Caddy Media Gallery

Operator Documentation



A Caddy v2 module for serving image and video galleries with a unified lightbox, configurable sort/pagination, and a templatable UI.

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Configuration

Caddyfile directive

The production setup — Caddy running as a system service, serving a site via HTTPS:

```
1 handle_path /images/* {
2     root * /var/www/html/images
3     media_gallery
4     file_server
5 }
```

For a local-dev setup (no sudo, no TLS), see the “Local install” subsection below — the Caddyfile is much simpler (just `http://localhost:3245` instead of a TLS hostname, no `admin` block needed for the bundled Caddyfile).

The `media_gallery` directive accepts one inline option:

Subdirective	Value	Default	Purpose
template	file name, relative to the templates dir	gallery.tmpl	Pick which template file to render. Path-traversal protected: no <code>..</code> , no absolute paths — the templates dir is a chroot.
path_prefix	URL prefix (e.g. <code>images</code>)	directory basename	URL mount prefix used in breadcrumb links. Defaults to the basename of the root directory if not set.
root_name	display name	media root	Display name for the root breadcrumb segment.

Subdirective	Value	Default	Purpose
image_types	space-separated extensions (no leading dot)	jpg jpeg png gif webp	File extensions the gallery treats as images. Case-insensitive. HEIC, AVIF, and SVG are NOT in the defaults — Go’s stdlib can’t decode them. Files with those extensions are classified as “other” files (in the “Other files” section, not the image grid) and shown with a  icon. Operators who want to handle these formats (e.g. via external tooling) can opt in with image_types .heic .avif .svg.
video_types	space-separated extensions (no leading dot)	mp4 webm m4v mov mkv avi ogv ogg	File extensions the gallery treats as videos.
sort	mtime / name	mtime	Sort field for the image grid. mtime = newest first; name = case-insensitive alphabetical.
page_size	integer >= 1	60 (or first item in page_sizes)	Default per-page count. Deprecated: use page_sizes (list form) for the dropdown, which lets the visitor choose.
page_sizes	space-separated list (first = default)	60 30 120 all	Per-page dropdown options. The first item is the default. Use the all token to include “show everything on one page” in the dropdown.
thumb_width	integer >= 1	320	Max width (px) of generated thumbnails.
thumb_height	integer >= 1	320	Max height (px) of generated thumbnails.

Subdirective	Value	Default	Purpose
thumb_format	webp / png / jpeg (or jpg)	webp	Output format for generated thumbnails.
thumb_ttl	integer (minutes) >= 1	1440 (24h)	HTTP Cache-Control: max-age for thumb responses.
cache_scan	integer (minutes) >= 1	1440 (24h)	In-memory scan cache TTL. The scan cache's primary invalidation is the directory mtime (checked on every access — adding or removing a file changes the dir mtime and invalidates the cache). The TTL is a safety net for edge cases (clock skew, manual mtime changes, stat cache invalidation). The default 24h keeps the cache warm for typical interactive use without forcing periodic re-scans.
no_thumbs	true / false (no-arg = true)	false (thumbs on)	Skip on-the-fly WebP thumbnail generation for images. Tile points to the original file instead of ~/_thumbs /<name>.webp. Thumb requests fall through to the next handler.
no_video_thumbs	true / false (no-arg = true)	false (video thumbs on, if ffmpeg available)	Skip ffmpeg-based video poster extraction.

Subdirective	Value	Default	Purpose
no_exif	true / false (no-arg = true)	false (EXIF on)	Disable EXIF reading entirely. When set, the visible-page sync enrich skips EXIF reads, the per-thumb serveThumb does not create .exif sidecars, and the lightbox EXIF panel is hidden (cards no longer show the “EXIF” pill). The dimensions watermark still appears (unaffected by no_exif). Useful for privacy-sensitive deployments. Note that EXIF does NOT include GPS by default — see the EXIF section for details.
search_match	word / substring	substring	Filename match rule for the search feature. word = match the start of a word boundary (the original Phase 118 behaviour). substring = match anywhere in the filename. Both server-side and client-side filters use the same rule.
max_cache_size_mb	integer >= 0	1024 (1 GB)	Cap on the on-disk thumb cache in MB. When the cache exceeds this, the oldest thumbs (by file mtime) are evicted until the cache is at 80% of the cap. Set to 0 to disable the cap entirely (unbounded — the pre-feature behavior). See Caching & performance below for the full story.

Example with a themed subdir:

```

1 handle_path /images/* {
2     root * /var/www/html/images
3     media_gallery {
4         template themes/dark/gallery.tmpl
5     }
6     file_server
7 }

```

This loads `$GALLERY_TEMPLATES_DIR/themes/dark/gallery.tmpl` (or falls back to the bundled template if the file doesn't exist on disk). The path is validated at Provision — an invalid name (e.g. `../etc/passwd`) fails Caddy startup, not at first request.

Local install (no root, no sudo)

For users without sudo access (shared host, locked-down laptop), use `./build.sh --user:`

```

1 ./build.sh --user          # default port 3245 (= 0xCAD, easter egg), serve ~/Pictures
2 ./build.sh --user 9000    # custom port (must be > 1024)
3 CADDY_USER_ROOT=~/.photos ./build.sh --user

```

This builds the binary to `~/bin/caddy` and generates `Caddyfile.user` in the project root:

```

1 {
2     admin off
3 }
4
5 http://localhost:3245 {
6     root * /home/user/Pictures
7
8     handle_path /* {
9         media_gallery
10        file_server
11    }
12 }

```

Then run:

```

1 ~/bin/caddy run --config Caddyfile.user
2 # or backgrounded:
3 nohup ~/bin/caddy run --config Caddyfile.user > ~/caddy.log 2>&1 &

```

The `--user` mode enforces the port (`>1024`, so it never needs root), uses the bundled template automatically (no `GALLERY_TEMPLATES_DIR` needed), and leaves any existing `Caddyfile.user` alone on subsequent builds.

The full argument matrix is in `./build.sh --help`.

Example: skip thumbnail generation

```

1 handle_path /images/* {
2     root * /var/www/html/images
3     media_gallery {
4         no_thumbs

```



```

5     }
6     file_server
7 }

```

With `no_thumbs`: - Each tile's `<img src>` is the original image file (`./photo.jpg`), not `~/_thumbs/photo.webp` - No thumb generation, no cache, no CPU cost on first request - The browser downloads the full image per tile (bigger page payload, slower on dirs of large photos)

Use `no_thumbs false` to turn it back on (the default is off, so the directive is opt-in).

Example: disable video thumbnails

Video thumbnails (per the table above) require `ffmpeg`. If `ffmpeg` isn't available, the gallery falls back to the placeholder gradient + play button automatically — you don't need to do anything. If `ffmpeg` IS available and you want to skip frame extraction (e.g., on a low-CPU host or for very large videos), use `no_video_thumbs`:

```

1 media_gallery {
2     no_video_thumbs
3 }

```

With this, video tiles show the placeholder gradient + play button (no `<img>` for the frame). The same `no_video_thumbs false` form re-enables frame extraction.

The video thumb generation uses `ffmpeg -vframes 1` to extract the first frame, scaled to fit the configured `thumb_width × thumb_height` (defaults 320×320). The output is a WebP, written to the same cache dir as image thumbs (`/var/cache/caddy-gallery` by default, override via `GALLERY_THUMB_CACHE_DIR`). Same caching rules as image thumbs (regenerate only when the source video's mtime is newer than the cache file).

If the operator has `ffmpeg` installed at a non-standard path (not in `$PATH`), set the `FFMPEG_PATH` env var to the absolute path of the `ffmpeg` binary. This is checked first (Phase 67); if unset or the file isn't executable, the code falls back to `exec.LookPath("ffmpeg")` which scans `$PATH`.

```

1 FFMPEG_PATH=/opt/ffmpeg-7/bin/ffmpeg caddy run

```

The `FFMPEG_PATH` value is validated at Provision time: it must point to an existing regular file with at least one executable bit set. Bad values (non-existent path, directory, non-executable file) are silently ignored and the code falls back to `$PATH` lookup. This avoids a confusing “exec: not found” error at request time when the env var is mistyped.

All standard install paths (`/usr/bin/ffmpeg`, `/usr/local/bin/ffmpeg`, etc.) are picked up automatically via the `$PATH` fallback.

`ffmpeg` detection is startup-only — restart Caddy after installing it

`FFMPEG_PATH` and the `$PATH` lookup are evaluated exactly once, in the Caddy module's `Provision()` (the startup hook). The resolved path is cached on the `Gallery` struct and reused for every video thumbnail request thereafter. There is no per-request re-scan of `$PATH`, no per-request stat of the binary, and no hot-reload if `ffmpeg` is installed or upgraded while Caddy is running.

If you install `ffmpeg` after Caddy is already running, video thumbnails will continue to fail (404 from the thumb handler) until you restart Caddy. The restart re-runs `Provision()`, which re-evaluates `FFMPEG_PATH` and `$PATH` and re-caches the resolved path. The same applies if you change `FFMPEG_PATH` — it only takes effect on the next startup.

This is deliberate: the alternative (per-request detection) would mean inconsistent behavior during the transition window — newly-installed `ffmpeg` would work for some requests but not others, depending on timing — and would add an `exec.LookPath` syscall to every video thumb request for no real benefit.

To pick up a new or upgraded `ffmpeg`:

```
1 # System install
2 sudo systemctl restart caddy
3
4 # Local install (--user)
5 kill $(cat ~/caddy.pid) && ~/bin/caddy run --config Caddyfile.user
```

The Caddy startup log shows the resolved `ffmpeg` path (or a warning if it wasn't found) so you can verify the right binary was picked up — see the `Provision()` log output when Caddy starts.

Best for: small galleries (< 100 images) where you don't want a thumb cache and the originals aren't huge. Not recommended for large galleries — the page payload goes from ~30 KB (with thumbs) to ~5 MB average (full images) for a 1,000-image dir.

Example: change the page size

```
1 handle_path /images/* {
2     root * /var/www/html/images
3     media_gallery {
4         page_size 100
5     }
6     file_server
7 }
```

This shows 100 image entries per page instead of the default 60. Tradeoffs: larger pages mean fewer HTTP requests, but each request returns a bigger HTML payload (and the server uses more memory per render). The pagination nav at the bottom of the page only renders when total pages > 1, so if your gallery has 30 images and you set `page_size 100`, you get all 30 on one page with no nav. (You can also use `page_sizes 100` to expose 100 in the dropdown and let the visitor switch back to 60.) URL query override: append `?page=2` to the gallery URL to jump to a specific page. `?page_size=N` is a valid URL query param: the visitor can switch the per-page size via the dropdown in the meta line. The value is validated against the operator-configured `page_sizes` list (an unknown value falls back to the first item). Changing the page size resets the visitor to page 1 (so they don't end up on a non-existent page).

All other configuration (the `root *` for the image directory, the `handle / handle_path` for the route, the auth wrapper) is via the surrounding Caddyfile block, not the `media_gallery` directive. The module reads the on-disk directory from Caddy's per-request `root` variable, or from the `Root` JSON field if set explicitly.

JSON config (advanced)

Caddy supports two configuration formats: the Caddyfile (text, what most examples on this page use) and JSON (the native config format Caddy uses internally). Every Caddyfile gets converted to JSON before being applied — but you can also write JSON directly, which is useful for:

- Programmatic / templated config (Kubernetes, Terraform, Ansible) — JSON can be generated from variables

- Many Caddy instances — JSON is diffable, lintable, and can be validated in CI pipelines
- Dynamic reload — `caddy reload` accepts JSON via the admin API (`curl -X POST http://localhost:2019/load`)
- Sharing snippets — unambiguous quoting (no special-character rules like Caddyfile)

Minimum JSON config

The minimum handler block (only `handler` is required):

```
1 {
2   "handler": "media_gallery",
3   "root": "/var/www/html/images"
4 }
```

The `Root` field is optional — the module falls back to the per-request root set by the surrounding `handle` / `handle_path` block. Set it explicitly only if you need to override the request-time root.

Full JSON config (all fields)

Here's a complete JSON config showing every configurable field of the `media_gallery` handler, with realistic values:

```
1 {
2   "handler": "media_gallery",
3   "root": "/var/www/html/images",
4   "path_prefix": "images",
5   "root_name": "images",
6   "image_types": ["jpg", "jpeg", "png", "gif", "webp"],
7   "video_types": ["mp4", "webm", "m4v", "mov", "mkv", "avi", "ogv", "ogg"],
8   "sort": "name",
9   "page_size": 60,
10  "page_sizes": ["60", "30", "120", "all"],
11  "thumb_width": 320,
12  "thumb_height": 320,
13  "thumb_format": "webp",
14  "thumb_ttl": 1440,
15  "cache_scan": 1440, // 24h (mtime check is the primary invalidation)
16  "no_thumbs": false,
17  "no_video_thumbs": false,
18  "template": "gallery.tmpl",
19  "search_match": "substring",
20  "max_cache_size_mb": 1024
21 }
```

All fields are optional except `handler` (always required). Defaults match the Caddyfile defaults — if you omit a field, the same default value applies.

Caddyfile ↔ JSON field mapping

Caddyfile directive	JSON field	Type	Default
<code>path_prefix <prefix></code>	<code>"path_prefix"</code>	string	(directory basename)
<code>root_name <name></code>	<code>"root_name"</code>	string	media root

Caddyfile directive	JSON field	Type	Default
image_types <ext1 ext2 ...>	"image_types"	[]string	built-in list
video_types <ext1 ext2 ...>	"video_types"	[]string	built-in list
sort <mtime\ name>	"sort"	string	mtime
page_size <N>	"page_size"	int	(first item of page_sizes)
page_sizes <N1 N2 ...>	"page_sizes"	[]string	["60", "30", "120", "all"]
thumb_width <N>	"thumb_width"	int	320
thumb_height <N>	"thumb_height"	int	320
thumb_format <fmt>	"thumb_format"	string	webp
thumb_ttl <N>	"thumb_ttl"	int (minutes)	1440
cache_scan <N>	"cache_scan"	int (minutes)	1440 (24h)
no_thumbs / no_thumbs false	"no_thumbs"	bool	false
no_video_thumbs / no_video_thumbs false	"no_video_thumbs"	bool	false
template <name>	"template"	string	gallery.tmpl
search_match <word\ substring>	"search_match"	string	substring
max_cache_size_mb <N >	"max_cache_size_mb"	int (MB)	1024 (1 GB; 0 = no cap)
thumb_width <N>	"thumb_width"	int	320
thumb_height <N>	"thumb_height"	int	320
thumb_format <webp\ jpeg\ png>	"thumb_format"	string	"webp"
thumb_ttl_minutes <N >	"thumb_ttl"	int	1440 (24h)
cache_scan_minutes < N>	"cache_scan"	int	1440 (24h)

Heads up on the JSON naming: the JSON field names use the `json:"name"` struct tags — for `CacheScanMinutes` the tag is `cache_scan` (not `cache_scan_minutes`), and for `ThumbTTLMinutes` it's `thumb_ttl` (not `thumb_ttl_minutes`). This is intentional: the Go struct tags are short, and the Caddyfile subdirectives keep the verbose names. The mapping table above is authoritative for what JSON field names to use — the Go field names are an implementation detail.

The mapping for the other fields is mechanical: every Caddyfile subdirective has a matching JSON field with the same name (snake_case throughout). The Go struct field names are `Root`, `Sort`, etc. (PascalCase), but the JSON tags normalize them to snake_case via the `json:"name, omitempty"` struct tags.

Validation

You can validate a JSON config without starting Caddy:

```
1 caddy validate --config /etc/caddy/caddy.json
```

Output: Valid configuration (or a JSON parse error pointing to the problem). The `caddy validate` command works for both JSON and Caddyfile inputs — for Caddyfile, just point it

at the file directly.

Dynamic reload

Push a JSON config to a running Caddy via the admin API:

```
1 curl -X POST http://localhost:2019/load \  
2   -H "Content-Type: application/json" \  
3   --data-binary @new-config.json
```

Returns 200 OK on success, or a JSON error response on failure. This is what `caddy reload` does internally — but you can also script reloads (e.g., update the gallery config when the disk layout changes) by POSTing to this endpoint.

When to use JSON vs Caddyfile

For single-host, edit-by-hand use, Caddyfile is the right tool: simpler for humans, more forgiving of whitespace and comments, easier to read for someone not familiar with the schema. This is what most of this document shows.

For automated / multi-instance / programmatic use, JSON is the right tool: it's diffable, lintable, validatable, and can be generated from templates. This is what Caddy uses internally, and what most CI/CD pipelines produce.

You can also convert between them: `caddy adapt --config Caddyfile` produces JSON on stdout, and `caddy adapt --config caddy.json --adapter caddyfile` goes the other way (rare, since the Caddyfile syntax is less expressive).

Environment variables

The plugin reads one environment variable:

`GALLERY_TEMPLATES_DIR`

Where it's read in code: `render.go`, in two functions — `loadTemplate()` (at request time, to find the on-disk override) and `writeBundledTemplates()` (at Caddy startup, to write the bundled templates so operators can see them). Both do the same thing:

```
1 dir := os.Getenv("GALLERY_TEMPLATES_DIR")  
2 if dir == "" {  
3     dir = "/etc/caddy/gallery-templates"  
4 }
```

Default: `/etc/caddy/gallery-templates`. Created on first Caddy startup by `writeBundledTemplates()` with mode `0755`, owned by whatever user the Caddy systemd service runs as. If the template file already exists, it's left alone (operator overrides are preserved across restarts).

The directory it points to: the absolute path you set. The plugin does not create parent directories beyond the templates dir itself — it just `MkdirAll`s the final directory. So if you point it at `/srv/gallery-templates`, that path needs to be writable by the Caddy process.

How to set it for the live Caddy (the canonical way — systemd starts the process with a clean environment, so your shell's env doesn't reach the service):

```

1  # One-off (persists in the manager's env, inherited by all units)
2  sudo systemctl set-environment GALLERY_TEMPLATES_DIR=/etc/caddy/gallery-templates
3  sudo systemctl restart caddy
4
5  # Or persistently in the caddy service unit
6  sudo systemctl edit caddy
7  # Add:
8  # [Service]
9  #   Environment="GALLERY_TEMPLATES_DIR=/etc/caddy/gallery-templates"
10 # Or use EnvironmentFile= pointing at a file with the line
11 sudo systemctl daemon-reload
12 sudo systemctl restart caddy

```

How to set it for dev (go test, xcaddy build, or running a custom Caddy from your shell):

```

1  export GALLERY_TEMPLATES_DIR=/path
2  # then run your commands

```

Note: the test suite sets GALLERY_TEMPLATES_DIR to a temp dir via TestMain, so go test is isolated from your shell's value. The module always reads the env at request time, so changes to the env var take effect on the next Caddy restart — no rebuild needed.

What happens if the dir is unwritable or doesn't exist: writeBundledTemplates() logs a warning to stderr and continues. The bundled templates still serve fine (the on-disk file is a convenience, not a requirement). The next request falls back to the bundled constant.

There are no other env vars. Sort order, pagination, page size, and the thumb cache directory are all configurable in code (constants in the source) rather than via env vars.

In-code constants (not configurable)

These are baked into the source. They can be changed by editing the code, rebuilding, and restarting Caddy — but they're not exposed as Caddyfile directives or env vars. Listed here so you know what the defaults are and where to find them.

Constant	Value	Where
Thumbnail size	320px	thumbnails.go
Thumb Cache-Control max-age	86400 (24h)	thumbnails.go
Thumb format	WebP (lossless, via nativewebp)	thumbnails.go
Scan cache TTL	1 minute	gallery.go (NewScanCache(time.Minute))
Thumb width (px)	320	thumbnails.go (the resize)

Note: "Thumbnail size" and "Thumb width (px)" both reference the same constant (320) — the maximum dimension of the generated thumbnail. If you wanted them to be different values (say, max-dim 320 and width 280 for a non-square crop), let me know and I'll split them into two separate constants.

Why these aren't configurable yet: most operators never need to change them. The thumb size and Cache-Control max-age are sensible defaults that work for the vast majority of galleries. The scan cache TTL is short enough (1 minute) that new files appear quickly without

manual cache busting, and long enough that hot directories don't re-scan on every request. If you find yourself wanting to change one of these, that's a signal that the constant should probably be promoted to a config option — file an issue and we can discuss.

Caddy-level configuration (optional)

The `media_gallery` directive handles the rendering and on-the-fly thumbnailing. A few performance knobs live at the Caddy level (outside the module) and are recommended for any production deployment:

Compression

Caddy's built-in `encode` middleware compresses text responses (HTML, CSS, JS, JSON, SVG) with `gzip` and/or `zstd`. The gallery page is ~160 KB raw; `gzip` compresses it to ~20 KB (an 8x reduction), so first-paint and time-to-interactive drop significantly. Thumbnails are pre-compressed image formats (WebP), so the encoder leaves them alone.

```
1 route {
2     encode zstd gzip
3     ...
4 }
```

`zstd` is preferred by modern browsers (slightly smaller than `gzip` at similar decompression speed); `gzip` is the universal fallback. Caddy auto-selects the best encoding based on the `Accept-Encoding` header the browser sends.

The `encode` directive is built into Caddy core — no extra `--with` flag is needed at build time.

Static asset caching

`/favicon.ico`, theme CSS, and font files served by the gallery can have long `Cache-Control` max-age values to avoid re-downloading on every navigation:

```
1 header /favicon.ico Cache-Control "public, max-age=86400"
```

For per-page assets (thumbnails), the module sets its own TTL via the `thumb_ttl` subdirective (default 24h).

What `media_gallery` does NOT do

- It does not generate thumbnails at build time — thumbnails are generated on-the-fly on first request and cached in `/var/cache/caddy-gallery/`. See the wiki page for cache details.
- It does not handle uploads or modifications. It's read-only.
- It does not support nested directory pagination — when you click into a subdirectory, that subdirectory has its own gallery with its own pagination, but the parent dir's image list isn't merged in.
- It does not redirect from `/foo` to `/foo/`. If you request `/images/generated` (no trailing slash) and it's a directory, the module returns a 301 with a relative `Location: generated/` header (so the browser resolves it relative to the current URL). This is a workaround for Caddy's `handle_path` rewriting both `r.URL.Path` and `r.RequestURI`, which makes absolute reconstruction impossible inside the handler.

caddy_media_gallery — Complete configuration reference

A single-page reference for every configuration knob the `media_gallery` Caddy handler exposes. Use this as the “what can I configure” reference; for how to use individual features, see the per-topic docs linked below.

For per-topic deep dives: - `configuration.md` — Caddyfile directive details + env vars
- `templates.md` — the template index page (links to the per-topic pages: 03a-structure, 03b-customization, 03c-feature-dark-mode, 03d-feature-localstorage, 03e-feature-lightbox, 03f-feature-layout, 03g-building-pdf) - `sort-and-pagination.md` — the URL query API

1. Caddyfile subdirectives (the primary way)

Inside an `media_gallery { ... }` block:

Subdirective	Value	Default	Purpose
<code>path_prefix</code>	URL prefix (e.g. <code>images</code>)	directory basename	URL mount prefix used in breadcrumb links.
<code>root_name</code>	display name	<code>media root</code>	Display name for the root breadcrumb segment.
<code>image_types</code>	space-separated extensions (no leading dot)	built-in list	File extensions the gallery treats as images.
<code>video_types</code>	space-separated extensions (no leading dot)	built-in list	File extensions the gallery treats as videos.
<code>sort</code>	<code>mtime</code> / <code>name</code> (also accepts <code>date</code> as an alias for <code>mtime</code>)	<code>mtime</code> (newest first)	Default sort field. Overridable per-request via <code>?sort=</code> .
<code>page_size</code>	integer ≥ 1	(first item of <code>page_sizes</code>)	Default per-page count. Deprecated for new configs — use <code>page_sizes</code> (list form) instead.
<code>page_sizes</code>	space-separated list (first = default)	<code>60 30 120 all</code>	Per-page dropdown options. The first item is the default. Use the <code>all</code> token to include “show everything on one page” in the dropdown.
<code>thumb_width</code>	integer ≥ 1	<code>320</code>	Max width in pixels. Source is fit-within-bounds.
<code>thumb_height</code>	integer ≥ 1	<code>320</code>	Max height in pixels. Source is fit-within-bounds.

Subdirective	Value	Default	Purpose
thumb_format	jpeg / jpg / png / webp	webp (lossless)	Output format. jpeg quality 75, png lossless, webp lossless.
cache_scan	integer >= 1	1440 (24h)	Scan cache TTL in minutes. The primary invalidation is the directory mtime check on every access; the TTL is a safety net for edge cases.
thumb_ttl	integer >= 1	1440	HTTP Cache-Control: max-age for thumbs, in minutes (= 24h default).
no_thumbs	no-arg = true / explicit false = false	false (thumbs on)	Skip thumbnail generation. Tile points to the original file.
no_video_thumbs	no-arg = true / explicit false = false	false (video thumbs on, if ffmpeg available)	Skip ffmpeg-based video poster extraction.
no_exif	no-arg = true / explicit false = false	false (EXIF on)	Disable EXIF reading entirely. When set: the visible-page sync enrich skips EXIF reads, serveThumb does not create .exif sidecars, and the lightbox EXIF panel is hidden (cards no longer show the “EXIF” pill). The dimensions watermark is unaffected.
search_match	word / substring	substring	Privacy-friendly — useful when EXIF data is undesirable. Filename match rule for the search feature. word = match the start of a word boundary. substring = match anywhere.
template	file name, relative to the templates dir	gallery.tmpl	Which template file to render. Path-traversal protected.

Full Caddyfile example (every directive set):

```

1  handle_path /images/* {
2      root * /var/www/html/images
3      media_gallery {
4          path_prefix images
5          root_name images
6          image_types jpg jpeg png webp
7          video_types mp4 webm
8          sort name
9          page_sizes 30 60 120 all
10         thumb_width 480
11         thumb_height 320
12         thumb_format jpeg
13         cache_scan 1440 # 24h - mtime check is the primary invalidation
14         thumb_ttl 60
15         search_match word
16         max_cache_size_mb 1024
17         template themes/dark/gallery.tmpl
18     }
19     file_server
20 }

```

2. JSON config fields (programmatic)

Same fields, json tags. Use caddy adapt or any JSON config producer to set them:

```

1  {
2      "root": "/var/www/html/images",
3      "path_prefix": "images",
4      "root_name": "images",
5      "image_types": ["jpg", "jpeg", "png", "webp"],
6      "video_types": ["mp4", "webm"],
7      "sort": "name",
8      "page_size": 60,
9      "page_sizes": ["30", "60", "120", "all"],
10     "thumb_width": 480,
11     "thumb_height": 320,
12     "thumb_format": "jpeg",
13     "cache_scan": 1440,
14     "thumb_ttl": 60,
15     "no_thumbs": false,
16     "no_video_thumbs": false,
17     "search_match": "word",
18     "template": "themes/dark/gallery.tmpl"
19 }

```

cache is runtime-only (excluded from JSON via json:"-").

3. Per-request URL query parameters (no config needed)

Param	Values	Default	Effect
sort	mtime / name / type / size (also accepts date as an alias for mtime)	inherits from Caddyfile	Sort field
order	asc / desc	depends on sort	Sort direction
page	integer >= 1	1	Which page (only meaningful when there are > 1 pages)
page_size	any value in the operator-configured page_sizes list	first item	Visitor's per-page selection (driven by the dropdown). Changing this resets the visitor to page 1. Unknown values fall back to the first item.
q	free text (URL-encoded)	(none)	Server-side filename search. Combined with the visitor's search_match mode. Directories are never filtered.
type	comma-separated list of extensions (e.g. jpg, png), or the sentinel . for files without an extension	(none)	Server-side type filter. The form-submission version uses repeated ?ext=jpg&ext=png (both work). Use ext=. (literal dot) to filter to only files with no extension (e.g. Makefile, welcome).
dirs_sort / dirs_order	same as the main sort	name asc	Sort the Directories and Other Files tables. Header click is client-side (persists in data-search-match-style attributes + localStorage).

Deliberately NOT query-overrideable: - ?thumb_format=N — would let users force the server to recompute thumbs in any format - ?thumb_width=N / ?thumb_height=N — same

The thumb format and dimensions are set in the Caddyfile only. The page size IS now a query-overrideable URL param (via the visitor's dropdown), but the operator can only set it to values in their pre-configured page_sizes list.

4. Environment variables

Var	Default	Purpose
GALLERY_TEMPLATES_DIR	/etc/caddy/gallery-templates	Where the module looks for (and writes) the on-disk template override. The only user-facing env var.

A second env var exists in the code for testing only: GALLERY_THUMB_CACHE_DIR (default /var/cache/caddy-gallery) — not documented as a user-facing knob.

See configuration.md to Environment variables for the full setup story (systemd unit, dev workflow, failure mode).

5. Caddyfile handler-level wiring (external, not `media_gallery`-specific)

These live in the surrounding Caddyfile, not inside the `media_gallery { ... }` block:

Caddyfile construct	Purpose	Example
<code>handle_path</code> (or <code>route</code>)	Strips URL prefix and routes to <code>media_gallery</code>	<code>handle_path /images/* { ... }</code>
<code>root *</code>	Per-request “root” var, read by <code>media_gallery</code> to find the directory to scan	<code>root * /var/www/html/images</code>
<code>file_server</code>	Fallthrough for direct file requests and 404s (must come AFTER <code>media_gallery</code>)	<code>file_server</code>
Auth (Authelia, <code>basic_auth</code> , etc.)	Wraps the route in auth	(auth) snippet (or <code>basicauth { ... }</code> block)

6. External: the Caddyfile as a whole

The live site typically looks like:

```

1 hermes.synapticloop.com {
2     import auth
3     route /images/* {
4         root * /var/www/html/images
5         media_gallery
6         file_server
7     }
8 }
```

The (auth) snippet is defined elsewhere in the Caddyfile and wraps every route in Authelia `forward_auth` at 127.0.0.1:9091.

7. Behavior modifiers (not configurable, hardcoded)

These are baked into the source. They could be promoted to config options if a use case comes up, but currently aren't:

Constant	Value	Where
Thumb JPEG quality (when <code>thumb_format jpeg</code>)	75	<code>thumbnails.go</code>
Video thumb extraction	uses <code>ffmpeg -vframes 1</code> (first frame), scaled to <code>thumb_width × thumb_height</code>	<code>thumbnails.go</code>
Video thumb max retries	1 (no retry; failed extraction leaves the placeholder)	<code>thumbnails.go</code>
Lightbox	page-scoped (visits the current page only, not all pages in the directory)	<code>render.go</code> (lightbox JS in the template)
EXIF fields extracted	CAMERA subset only (Make, Model, Lens, <code>DateTimeOriginal</code> , <code>ExposureTime</code> , <code>FNumber</code> , <code>ISOSpeedRatings</code> , <code>FocalLength</code>). GPS data is never read.	<code>exif.go</code>
EXIF parser	<code>github.com/dsoprea/go-exif/v3</code> (supports JPEG, PNG, WebP)	<code>exif.go</code>
Search match mode default	<code>substring</code> (most permissive; the operator can opt into <code>word</code> via <code>search_match word</code>)	<code>exif.go + gallery.go</code> (Provision)
Dimensions reading	<code>image.DecodeConfig</code> (images, header-only) + <code>ffprobe</code> (videos, 10s timeout)	<code>dimensions.go</code>
Per-page dropdown default list	60 30 120 all (first item is the default)	<code>gallery.go</code> (Provision)
Page-size reset behaviour	changing the page size via the dropdown resets the visitor to page 1 (current <code>?page=</code> is dropped from the form's hidden inputs)	<code>render.go</code> (page-size form)
Other-files strip on a sub-dir page	“Other files” rendered as a table above the image grid, with click-to-sort headers	<code>render.go</code> (<code>splitFiles</code>)
Directories strip on every page	Always rendered, alphabetical, ignores sort. Has click-to-sort headers too (name / # items / # sub-dirs / size / modified).	<code>render.go</code> (<code>splitFiles + dirs sort</code>)
Lightbox EXIF panel	shows Camera, Lens, Date, Exposure (Shutter · Aperture · ISO · Focal) — only when the image has EXIF	<code>render.go</code> (lightbox JS)

Constant	Value	Where
Thumbnail loading indicator	subtle diagonal shimmer while the thumbnail image loads, removed on load / error events	<code>render.go</code> (inline JS)

8. Quick reference — what’s where

If you’re wondering “where is this knob?”:

Want to configure...	Use
Sort field (default + per-request)	<code>sort directive + ?sort= query</code>
Sort direction	<code>?order= query (asc / desc)</code>
Pagination	<code>page_size (default) + page_sizes (dropdown list) + ?page= query</code>
Per-page dropdown selection	<code>?page_size= query (validated against page_sizes list; resets to page 1 on change)</code>
Type filter (server-side)	<code>?type=jpg,png or repeated ?ext=jpg&ext=png</code>
Filename search (server-side)	<code>?q=foo (combined with the visitor’s search_match mode)</code>
Search match mode	<code>search_match directive (word or substring, default substring)</code>
Thumbnail on/off	<code>no_thumbs directive</code>
Video thumbnail on/off	<code>no_video_thumbs directive</code>
Thumbnail size	<code>thumb_width / thumb_height directives</code>
Thumbnail format (jpeg/png/webp)	<code>thumb_format directive</code>
Thumbnail browser cache TTL	<code>thumb_ttl directive</code>
Directory scan cache TTL	<code>cache_scan directive</code>
Image extensions	<code>image_types directive (space-separated)</code>
Video extensions	<code>video_types directive (space-separated)</code>
URL mount prefix (for breadcrumb)	<code>path_prefix directive</code>
Root breadcrumb display name	<code>root_name directive</code>
Template file (theme)	<code>template directive</code> (e.g. <code>themes/dark/gallery.tmpl</code>)
Templates directory (where to find templates)	<code>GALLERY_TEMPLATES_DIR env var</code>
Thumbnails cache directory	<code>GALLERY_THUMB_CACHE_DIR env var (testing only)</code>
EXIF parsing	automatic (no config). CAMERA fields only; GPS data NEVER read.
Source dimensions in thumbnail	automatic (no config). Bottom-left watermark on every card.
Auth (who can view)	Caddyfile <code>basicauth</code> or <code>(auth) snippet (external)</code>
Route (URL prefix)	Caddyfile <code>handle_path</code> or <code>route (external)</code>
Image directory	Caddyfile <code>root *</code> (external)

Templates

The gallery is rendered by an `html/template` (Go's standard template engine). The template is bundled in the binary as a Go string constant (`galleryTemplate` in `render.go`), and on each Caddy startup the module also writes it to disk at `/etc/caddy/gallery-templates/gallery.tpl` (or `$GALLERY_TEMPLATES_DIR` if set). The on-disk file exists so operators can read the template without opening the binary, and so they can override individual templates by editing the file in place.

This page is the index for the template docs. Pick a topic:

- [03a-structure.md](#) — How the template loads, what's in the file, and the variables it uses
- [03b-customization.md](#) — How to edit the templates (walkthroughs, troubleshooting, upgrading)
- [03c-feature-dark-mode.md](#) — Dark mode and the theme toggle
- [03d-feature-localstorage.md](#) — What the template stores in `localStorage`
- [03e-feature-lightbox.md](#) — The lightbox (image + video), its controls, the EXIF/META panels
- [03f-feature-layout.md](#) — Header, pagination, footer, section heading, open-in-new-tab button
- [03g-building-pdf.md](#) — Building the operator manual PDF locally

Templates: structure and variables

How the template loading works

```
1 loadTemplate(name string) reads $GALLERY_TEMPLATES_DIR
2
3   └─ name = gallery.Template (from Caddyfile `template` directive)
4     └─ default "gallery.tpl" if the directive is absent
5
6   └─ sanitizeTemplateName(name):
7     └─ reject absolute paths, reject ".." (any traversal)
8
9   └─ /etc/caddy/gallery-templates/<clean name> exists?
10      └─ YES to parse that file directly (CSS+JS are inside it)
11         └─ (the on-disk file is the active template)
12      └─ NO to use the bundled galleryTemplate constant
13
14   └─ return *template.Template ready to RenderPage
```

The template Caddyfile directive is the operator-facing knob that picks the template file. See [docs/configuration.md](#) for the directive syntax and the path-traversal protection details.

Note on `no_thumbs`: the template is unaffected by the `no_thumbs` Caddyfile directive. The template always uses `{{.ThumbURL}}` for the tile `<img src>`; the `no_thumbs` flag changes the *value* of that field (to the original file URL when true, to the thumb URL when false), not the field name or its usage. So a template that works with thumbs also works with `no_thumbs` (and vice versa), with no template changes needed.

The template is a single self-contained file. The HTML, CSS (inside `<style>`), and JS (inside `<script>`) all live in one Go string constant (`galleryTemplate` in `render.go`) and one on-disk file (`/etc/caddy/gallery-templates/gallery.tpl`). There is no sub-template loading — the previous

Phase 16 design that split them into 3 files (gallery.tpl + style.css + lightbox.js) was collapsed in Phase 17 for easier editing.

The bundled constant is the source of truth at build time. The on-disk file is an *override* layer; if you delete it, the module falls back to the bundled version automatically on the next request.

What's in the single template file

One file, gallery.tpl, ~16.7 KB / 574 lines, containing:

Section	What's in it	Approx lines
<!DOCTYPE> ... <style>	Document head, including the full CSS inlined as a <style> block	~10 + 365 (CSS)
<body>, <main>, <header>	Title, counts, sort indicator, sort bar	~50
Directories section	{{if .Directories}} ... {{end}} — the dirs chip strip	~7
Other files section	{{if .OtherFiles}} ... {{end}} — the others chip strip	~7
Images section	Sort info, paginated grid, per-tile HTML	~80
Pagination	{{if gt .TotalPages 1}} ... {{end}} — prev/next links	~15
<script> ... </script>	The full JS inlined (lightbox, open-in-new-tab, sort indicator)	~100 (JS)

The CSS rules and the HTML they apply to are interleaved top-to-bottom in the file, so when you scroll through the template you see the structure (HTML), the styling (CSS), and the behavior (JS) in document order. Easier to hand-edit than three separate files.

Template variables

The template receives a PageData struct. All fields you can reference from {{...}} are listed below.

Top-level fields (on `PageData`)

Field	Type	What it's for
{{.Title}}	string	The page title (rendered in <title> and the <h1>). Currently the directory name.
{{.PathPrefix}}	string	Prefix for relative links to files in the same directory. Usually "./".
{{.ThumbPrefix}}	string	Prefix for thumbnail URLs. Usually "./_thumbs/".

Field	Type	What it's for
{{.Up}}	*FileView	The synthetic “Up” entry for the parent directory. nil at the gallery root.
{{.Subdirs}}	[]FileView	Subdirectory entries. Always rendered in full (not paginated), case-insensitive alphabetical.
{{.OtherFiles}}	[]FileView	Non-media files (HTML, txt, etc.). Always rendered in full (not paginated).
{{.Images}}	[]FileView	The image + video tiles for the current page only. Paginated, sorted per the user's ?sort=&order= choice.
{{.Page}}	int	Current page number (1-based).
{{.PageSize}}	int	Current per-page count (driven by the dropdown / ?page_size= query).
{{.PageSizes}}	[]string	The visitor's per-page dropdown options (e.g. ["30", "60", "120", "all"]).
{{.ImageStart}}	int	1-based start of the current page's image range (e.g. 1 for page 1, 61 for page 2).
{{.ImageEnd}}	int	1-based end of the current page's image range (e.g. 60 for page 1 of 60-per-page, 89 for page 2 of 60-per-page of a 89-image dir).
{{.TotalImages}}	int	Total images in the directory (across all pages, after type/q filters).
{{.TotalPages}}	int	Total page count.
{{.TotalFiles}}	int	Total files in the directory (images + videos + other files).
{{.TotalVideos}}	int	Total video files in the directory.
{{.ImageCount}}	int	Count of image files only (videos excluded).
{{.TotalAllFilesSize}}	string	Human-readable total size of all files.
{{.HasPrev}}	bool	True if {{.Page}} > 1.
{{.HasNext}}	bool	True if {{.Page}} < {{.TotalPages}}.
{{.PageNumbers}}	[]int	Google-style page-number list for the pagination nav.
{{.Sort}}	SortSpec	May contain 0 (a “...” ellipsis). The current sort. See below.

Field	Type	What it's for
<code>{{.SearchQuery}}</code>	string	The raw <code>?q=</code> value (preserved across sort changes).
<code>{{.SearchMatch}}</code>	string	The operator-configured search match mode ("word" or "substring"). Used to render a <code>data-search-match</code> attribute on the search input so the inline JS uses the same rule as the server.
<code>{{.TypeFilter}}</code>	map[string]bool	The active type filter (e.g. <code>{"jpg": true, "png": true}</code> for <code>?type=jpg,png</code>).
<code>{{.IsTypeFilterActive}}</code>	bool	True if any type filter is active.
<code>{{.TypeFilterQuery}}</code>	string	The raw <code>?type=</code> value (the canonical, comma-separated form).
<code>{{.FilterImageOptions}}</code> / <code>{{.FilterVideoOptions}}</code> / <code>{{.FilterOtherOptions}}</code>	structs	Per-type filter groups (each option's extension, label, count, selected).
<code>{{.Breadcrumb}}</code>	[]BreadcrumbSegment	Breadcrumb trail from gallery root to current dir. Each segment has Name, Href, IsCurrent.
<code>{{.Query}}</code>	url.Values	Raw URL query — useful for the per-page form to preserve all params except the ones it overrides.

`{{.Sort}}` (a `SortSpec` struct)

Field	Type	What it's for
<code>{{.Sort.Field}}</code>	string	One of "mtime", "name", "type", "size". (The <code>?sort=</code> URL param. "mtime" is the default.)
<code>{{.Sort.Order}}</code>	string	"asc" or "desc". (The <code>?order=</code> URL param.)

Per-entry fields (on `FileView`)

When you `{{range .Images}}` (or `.Directories` / `.OtherFiles`), each iteration gives you a `FileView` with these fields:

Field	Type	What it's for
{{.Name}}	string	The basename ("photo.jpg", "subdir", etc.). Truncated with ellipsis in the live template if too long for the tile.
{{.Href}}	string	Relative link to the file. Use as <code></code> .
{{.ThumbURL}}	string	For images and videos, the relative thumbnail URL (e.g. <code>./_thumbs/photo.webp</code>). Empty string for non-media files — check <code>{{if .ThumbURL}}</code> before using.
{{.IsDir}}	bool	True for directories.
{{.IsImage}}	bool	True for image files.
{{.IsVideo}}	bool	True for video files. Videos go in the image grid with a play-button overlay (no <code></code> child); <code>{{.ThumbURL}}</code> is set but the live template doesn't render an <code></code> for them.
{{.IsOther}}	bool	True for non-media files (HTML, txt, etc.).
{{.Type}}	string	Uppercase extension without the dot: "JPG", "DIR", "MP4", "HTML", etc.
{{.Size}}	string	Human-readable file size: "234 KB", "1.2 MB", etc.
{{.Date}}	string	Empty string for directories. ISO date "2006-01-02" (UTC-normalised). Empty string for directories.

Template functions (the funcmap)

The template engine has a few helper functions registered:

Func	Signature	What it does
minus1	minus1 n int to int	Returns <code>n - 1</code> . Used for prev-page link targets.
plus1	plus1 n int to int	Returns <code>n + 1</code> . Used for next-page link targets.
lastIndex	lastIndex s []T to int	Returns <code>len(s) - 1</code> . Used for the breadcrumb.

Func	Signature	What it does
sortLabel	sortLabel field string to string	Maps a sort field code to its display label: "name"to"Name", "type"to"Type", "mtime"to"Modified", "size"to"Size", "date"to"Date". Unknown fields fall back to the raw field name capitalised. Empty string to "Modified" (the default).
queryString	queryString q url.Values to template.URL	Renders url.Values as a key=val&key=val query string. Used for building pagination / sort / filter links. Returns empty string if no params.
queryWith	queryWith q url.Values, key, value string to url.Values	Returns a new url.Values with the given key replaced (or removed if value is empty).
queryForPage	queryForPage q url.Values, sort SortSpec, page int to url.Values	Pagination-specific: keeps the effective sort/order, replaces page.
sortURL	sortURL q url.Values, field, order string to url.Values	Sort-toggle-specific: sets sort + order, resets page to 1.
sortOrder	sortOrder currentField, field, currentOrder string to string	Returns the toggled order (asc [] desc if same field, else the new field's default).
queryToHiddenInputs	queryToHiddenInputs q url.Values to template.HTML	Renders url.Values as hidden <input> elements. Used by the per-page form to preserve other params on submit.
queryToHiddenInputsExclude	queryToHiddenInputsExclude q url.Values, exclude ... string to template.HTML	Like above, but with a variadic list of keys to exclude. Used by the per-page form to omit page (so changing size resets to page 1) and page_size (the dropdown supplies it).
formatDimensions	formatDimensions w, h int to string	Returns "WIDTH × HEIGHT" (e.g. "1024 × 1024") or empty if either is 0.

Templates: editing and customization

Editing the templates — the basics

1. Find the templates dir. On this host it's /etc/caddy/gallery-templates/. The file is gallery.tpl.
2. Edit the file in place. The module re-reads it on every request, so changes take effect immediately — no Caddy restart needed.

3. Test in a browser. The next request will use the new template. If the template has a parse error, the module will return 500 with the parser error in the response body. Fix the syntax and try again.
4. Revert to the bundled version. If you want to go back to what the binary ships with, delete the on-disk file: `sudo rm /etc/caddy/gallery-templates/gallery.tmpl`. The next request falls back to the bundled constant.

Walkthrough: change the theme to dark mode

Since the CSS is now inlined in the same file as the HTML, you edit the `<style>` block directly in `gallery.tmpl` — no second file to coordinate, no sub-template indirection.

1. Open `/etc/caddy/gallery-templates/gallery.tmpl` in your editor.
2. Find the `<style>` block. It's the big CSS block near the top of the file, just after `<title>...</title>`. Search for `html, body { background: #f3f6f7;` to find the body color.
3. Change the colors inline. The CSS is plain CSS — no preprocessing. Most of the theme colors are concentrated in the first 50-100 lines.
4. Save the file. The change takes effect on the next request (no Caddy restart needed — the module re-reads the on-disk template on every `loadTemplate` call).
5. Hard-reload in the browser to bypass the HTML cache.

A common set of swaps for dark mode (find/replace these in the `<style>` block):

Light (default)	Dark variant
<code>html, body { background: #f3f6f7; }</code>	<code>html, body { background: #1a1a1a; color: #ddd; }</code>
<code>main { background: white; }</code>	<code>main { background: #222; }</code>
<code>header { border-bottom: 1px solid #e5e9ea; }</code>	<code>header { border-bottom: 1px solid #333; }</code>
<code>.chip { background: #f3f6f7; border: 1px solid #e5e9ea; }</code>	<code>.chip { background: #2a2a2a; border: 1px solid #3a3a3a; }</code>
<code>a { color: #006ed3; }</code>	<code>a { color: #4ea3ff; }</code>

Walkthrough: add a “Created” column to the image tiles

If you want to show the file's created time alongside the modified date (the template currently shows Date which is the modified date):

1. Note: `FileView.Date` is the only date field exposed. The `FileInfo` struct in the scanner *does* have `ModTime` (in nanoseconds) but no `CreatedTime`. Adding a “Created” column would require a code change — extend `FileView` with a `Created` string field, populate it in `RenderPage` from `info.ModTime` (or from `os.Stat birthtime` via `syscall.Statx` on Linux), then reference it in the template as `{{.Created}}`.

In other words: this is a code change, not a template-only change. The template is fully driven by the Go struct fields — anything you want to display has to be in the struct.

Where the bundled templates are defined in source

If you want to read the source (or fork and customise):

File	Constant	Lines (approx)
<code>render.go</code>	<code>galleryTemplate</code>	line 392, ~574 lines (HTML + inlined CSS + inlined JS)

A single constant. CSS and JS are inlined inside `<style>` and `<script>` blocks respectively. To customise: edit the constant, rebuild the module (`./build.sh`), restart Caddy (`sudo systemctl restart caddy`). The on-disk template is written from the new constant on the next startup.

For local-install (`./build.sh --user`, no `sudo`), there's no `systemd` to restart — just kill the old caddy run process (`kill $(cat ~/caddy.pid)`) and start the new one. The bundled template is used automatically when no `Caddyfile.user` template subdirective is set, so there's nothing extra to do for the local case.

Upgrading from a pre-inlining install (Phase 16 to Phase 17)

If your site was running the old 3-file template split (`gallery.tpl` + `style.css` + `lightbox.js`) and you upgraded to the new inlining build, the on-disk files are in an inconsistent state:

- `style.css` and `lightbox.js` from the old build are now dead weight. `writeBundledTemplates` removes them automatically on the next Provision after upgrade. Safe to ignore.
- The on-disk `gallery.tpl` from the old build still has the old `{{template "lightbox.js" .}}` references, which no longer work. The new `loadTemplate` will fail to parse this file and the gallery will 500.

One-time fix on upgrade:

```
1 sudo rm /etc/caddy/gallery-templates/gallery.tpl
2 sudo systemctl restart caddy
```

The next Provision writes the new inlined template. After that, the on-disk file is the canonical inlined version, and any operator edits to it become live overrides.

Upgrading the bundled template content

`writeBundledTemplates()` deliberately does NOT overwrite an existing `gallery.tpl` on disk — that's the operator-override contract. When the bundled template's contents change (e.g. new CSS rule, new template branch, fixed layout bug), the on-disk file is NOT updated automatically.

To pick up the new bundled content:

```
1 # 1. Save any local customisations (if you have any)
2 diff /etc/caddy/gallery-templates/gallery.tpl /home/osmanj/projects/
   caddy_media_gallery/render.go
3 # 2. Delete the on-disk file
4 sudo rm /etc/caddy/gallery-templates/gallery.tpl
5 # 3. Restart Caddy so the next Provision runs writeBundledTemplates
6 sudo systemctl restart caddy
7 # 4. (Optional) Re-apply your local customisations
```

This workflow is intentional: operators who have customised the template (e.g. themed dark mode) keep their changes across upgrades. Operators who haven't customised just get the “fresh bundled version” after the `rm` + restart.

Future enhancement (not v1): `writeBundledTemplates` could write a content hash into a sidecar file, and on Provision, if the sidecar hash doesn't match the bundled hash, overwrite the on-disk file. This would auto-update without breaking the operator-override contract (the operator could still delete the on-disk file to fall back to the bundled version, OR write a sidecar with a custom hash to pin to their version).

Troubleshooting

Edit took effect but the page looks the same. Hard-reload (Cmd-Shift-R / Ctrl-F5). The browser may have cached the previous HTML.

Edit took effect but the page is a 500. Your template has a parse error. The response body will contain the Go `html/template` parser error. Common causes: - Unclosed `{{if}}` / `{{range}}` / `{{with}}` blocks - `{{end}}` mismatch (most common — Go templates require `{{end}}` to close every block) - Calling a method that doesn't exist on the data type (e.g. `{{.Foo.Bar}}` when `Foo` is a string) - An unescaped backtick inside a Go template comparison (backticks terminate the Go raw string the constant is in)

Edit took effect but layout is broken. You removed or re-ordered a structural element. The CSS selectors and the JS `querySelector` calls expect a specific DOM shape — search for the class name in the template to see what's expected.

Want to test a new template before deploying it. Stage the edit in a new file at, say, `/etc/caddy/gallery-templates/gallery.tmpl.staging`, then copy it over the live file once you're happy. Or `curl` the gallery to see the live HTML and check it with a browser inspector before saving.

Got a 500 immediately after upgrading. You have the pre-inlining on-disk `gallery.tmpl` from a previous build. See the “Upgrading from a pre-inlining install” section above.

Templates: dark mode and theme toggle

Dark mode + theme toggle

The template supports three themes:

1. Auto (default) — follows the visitor's OS preference via the `prefers-color-scheme` CSS media query. macOS, Windows, iOS, and Android all expose this preference; if the visitor's OS is in dark mode, the gallery renders in dark mode automatically.
2. Light — forces light mode regardless of OS preference.
3. Dark — forces dark mode regardless of OS preference.

The choice is presented as a small toggle button group in the page header. Three buttons: a gear icon (Auto), a sun icon (Light), a moon icon (Dark). The current state is marked with `aria-pressed="true"` and a slightly inset background.

Position (Phase 50/51): the toggle was originally at the top-LEFT of the header (just to the left of the page title). As of Phase 50, it sits at the top-RIGHT, where the old sort-order indicator used to be. The sort indicator itself was removed in Phase 50 (the sort UI still exists as the sort-bar below the header — the active sort field is still visible there).

Mobile layout (Phase 50/51): on screens ≥ 600 px wide, the header stacks vertically via `@media (max-width: 600px) { .header-top { flex-direction: column; }`. The visual order on mobile is:

1. Theme toggle (top, right-aligned via `align-self: flex-end`)
2. h1 + meta line (below, full width)

This is achieved with CSS `order` on the flex children, not by reordering the HTML source (the source order stays `h1` `meta` `toggle`, which is better for screen readers):

```
1 .header-main { order: 2; }
2 .theme-toggle { order: 1; align-self: flex-end; }
```

The toggle’s row order (`order: 1`) puts it above the header content (`order: 2`) on mobile; the `align-self: flex-end` keeps it right-aligned. On desktop (the default), neither `order` nor `flex-direction` applies, so the layout reverts to the normal horizontal flex with the toggle at the right (via the `justify-content: space-between` on `.header-top`).

The choice persists across visits via `localStorage` under the key `gallery-theme`. Values stored: `auto` | `light` | `dark`. `auto` is the default (no attribute set on `<html>`); `light` and `dark` set the `data-theme` attribute on `<html>`.

No flash of wrong theme

There’s a tiny inline `<script>` in the `<head>` that reads `localStorage` and applies the `data-theme` attribute to `<html>` BEFORE the body paints. This runs synchronously during HTML parsing, so the CSS already sees the right theme when the body renders. No flash of light theme when the visitor has chosen dark.

How dark mode is implemented

All colors are defined as CSS custom properties (CSS variables) in the `:root` selector at the top of the template’s `<style>` block. There are ~16 tokens, divided into semantic groups:

Group	Tokens
Backgrounds	<code>--bg</code> (page), <code>--bg-card</code> (cards), <code>--bg-chip</code> (chips), <code>--bg-hover</code> (chip hover), <code>--bg-active</code> (active chip)
Text	<code>--fg</code> (primary), <code>--fg-muted</code> (secondary), <code>--fg-faint</code> (tertiary), <code>--fg-disabled</code> (disabled)
Borders & shadows	<code>--border</code> , <code>--border-strong</code> , <code>--shadow</code> , <code>--shadow-strong</code>
Accents	<code>--accent</code> (links + borders), <code>--accent-hover</code> , <code>--accent-bg</code> (button fills — separate from <code>--accent</code> so the dark-mode button can be a muted darker blue without dimming link text)

The dark mode override is just a second block of token assignments that applies when:

- `@media (prefers-color-scheme: dark) { :root:not([data-theme="light"]) { ... } }` — the OS is in dark mode AND the user hasn’t explicitly picked light. The `:not([data-theme="light"])` selector is what makes the “Auto” state work: when the OS is dark but the user picked Light, the dark tokens don’t apply.
- `[data-theme="dark"] { ... }` — manual dark mode regardless of OS preference. Triggered by clicking the moon icon; choice persists in `localStorage`.

Why `--accent` and `--accent-bg` are separate tokens: the accent color serves two visual roles — text/borders (good as a bright color on a dark bg, e.g. `#4dabff`) and button fills (looks glaring as a bright color, should be muted, e.g. `#3b6fb6`). Splitting them lets each role have its own dark-mode value.

Dark mode refinements (Phase 40–42)

The first dark-mode pass had two issues:

- Chip bg too light (Phase 40): the chips (directories, other files, sort buttons, page buttons) had a `--bg-chip` value (`#1d1d1d` in dark mode) that was visibly lighter than the page bg (`#1a1a1a`), making the chips stand out as bright blobs. Fixed by setting `--bg-chip: #1a1a1a` (same as `--bg`) in dark mode, so chips blend in with the page; only the border + text show. This mirrors the light-mode behavior (where `--bg-chip` already equals `--bg`).
- Hardcoded colors in 11 CSS rules (Phase 41): a regex-based refactor (the original dark-mode implementation) caught 18 rules but missed 11 more that used hardcoded light-mode hex values. After a comprehensive sweep, all CSS rules now use `var()` token references. The only `#hex` colors left in the on-disk template are: the token definitions themselves (light + dark overrides), the video tile placeholder gradient (theme-independent), and the play button colors (white-on-dark, theme-independent).

Customizing colors

To override a color, edit the on-disk template at `/etc/caddy/gallery-templates/gallery.tmpl` and change the token value in the `:root` block. For example, to make the accent color green instead of blue:

```
1 :root {  
2   --accent: #2e8b57;      /* was #006ed3 */  
3   --accent-hover: #3aa86a; /* was #0095e4 */  
4   --accent-bg: #2e8b57;   /* was #006ed3 — used by active sort/page buttons */  
5 }
```

To add a new dark-mode-specific color, define it in all three blocks (`:root`, the media query, and `[data-theme="dark"]`).

Lightbox (always dark)

The lightbox overlay (the full-screen image/video viewer) has its own dark colors that are theme-independent — the dark background and white controls work in both modes. This is intentional: a dark overlay focuses attention on the content regardless of the page theme.

Templates: localStorage keys

What the template stores in localStorage

Per user request 2026-07-01: this section is the canonical reference for every key the template writes to `localStorage`. Use this when debugging “why is the visitor seeing X?” or when adding new persisted state to the template.

All keys are namespaced with the `gallery-` prefix so they don’t collide with the host site’s `localStorage` usage (the module is mounted under `/images/*`, but the visitor may have already visited other parts of the host site with their own `localStorage` keys).

Key	Values	Source	Read on	Written on
gallery-theme	"light" "dark"	Theme toggle in header	Inline script in <head> (before body paint, so no theme flash) — see Dark mode + theme toggle	Click on theme toggle button
gallery-dirs-sort	"name" (default) "modified" (server-side default — operator-configurable)	Directories table header click	Dirs-table sort script — see Header meta line format (the sort bar lives in the header)	Click on a dirs-table column header
gallery-dirs-order	"asc" "desc"	Directories table header click	Same	Same
gallery-others-sort	"name" "modified" (default)	Other-files table header click	Same	Same
gallery-others-order	"asc" "desc"	Other-files table header click	Same	Same
gallery-section-<section>	"collapsed"	The [-]/[+] toggle on the directories / other-files section headings	Section-toggle script (eager: on page load, the script reads all gallery-section-* keys and applies the collapsed class)	Click on a section heading (or the small toggle button)
gallery-lb-exif-collapsed	"true" (absent = expanded)	Click on the EXIF panel header in the lightbox	Lightbox JS — when opening, if the key is "true" the panel starts collapsed	Click on the EXIF panel header to collapse (key removed on expand)
gallery-lb-video-meta-collapsed	"true" (absent = expanded)	Click on the META panel header in the lightbox (videos only)	Lightbox JS — same pattern as EXIF	Same

<section> is one of dirs or others — same values used in the data-section="..." attribute on each section header.

Why localStorage (not URL query) for these states?

- Theme: personal preference, not bookmarkable intent. Would pollute shared links.
- Table sort state: same reasoning — the visitor wants to remember *their* preferred sort, but a shared link to the gallery shouldn't force the recipient into a particular sort. The URL still carries the sort (so a NEW visitor with no localStorage gets the URL's sort), but the visitor's own choice persists across reloads in localStorage.
- Section toggle: pure UI preference; bookmarkable intent would be confusing ("why doesn't this link show the directories?").

For all three: localStorage = persistent + per-visitor + no URL noise. URL is the “first-time canonical” path; localStorage is the “I’ve been here before” path.

Why the `gallery-` prefix matters

If you fork this template, change the prefix to your own namespace (e.g. `myphotos-`). Otherwise, two galleries hosted on the same hostname would share theme/sort/toggle state in unexpected ways (the visitor picks “dark” on site A, and their next visit to site B is also dark).

This is also why the template NEVER reads localStorage keys without the prefix — a `localStorage.getItem('theme')` would be a security/UX bug.

Dirs / Others table sorting

Each of the two tables (directories and other files) has its own sort state. The state resolution is:

1. URL query (`?dirs_sort=name&dirs_order=asc`) — the canonical state. A shared link forces the sort on first visit.
2. localStorage (`gallery-dirs-sort + gallery-dirs-order`) — the visitor’s remembered preference.
3. Server-side default — operator-configurable in the template (the JS reads `data-default-sort / data-default-order` attributes on the table headers).

On click: - Same column ☐ toggle direction (`asc` ☐ `desc`, `desc` ☐ `asc`) - Different column ☐ use that column’s `data-default-order` - Sort client-side (no server round trip — read rows once, append in new order) - Save to localStorage - Update URL via `history.replaceState` so the back button still works (no history pollution, just a silent URL replacement)

The sort fires immediately and updates the table within a single frame. No page reload, no spinner.

Section toggle (the `[-]` / `[+]` button)

The directories table and the other-files table each have a collapse-toggle button. Per Phase 71/74: clicking the WHOLE section heading (the `<h2>` row) toggles the state — the small button is still rendered for keyboard/screen-reader users.

The state is stored under `gallery-section-<section>` where `<section>` is the value of the `data-section` attribute on the `<h2>` (e.g. `dirs`, `others`). Value `"collapsed"` means collapsed; absence of the key means expanded (default).

Why localStorage for this and not URL state? A URL query would mean: shared links always force-collapse the dirs section. That’s annoying — the visitor can’t share a link that’s “the dirs section expanded”. localStorage is “remember MY last choice” with no URL pollution.

Clearing the template’s localStorage

To reset the visitor’s preferences (useful during operator testing):

```
1 [
2   "gallery-theme",
3   "gallery-dirs-sort", "gallery-dirs-order",
4   "gallery-others-sort", "gallery-others-order",
5   "gallery-section-dirs", "gallery-section-others",
6 ].forEach(k => localStorage.removeItem(k));
```

Or via DevTools: Application > Storage > Clear site data.

The next page load after clearing will use all server-side defaults — light theme (or auto-follow), name-asc dirs sort, date-desc others sort, both sections expanded.

localStorage unavailable?

Every read/write site has a `try { ... } catch (e) {}` wrapper. If localStorage is disabled (private/incognito mode in some browsers, disabled cookies/site-data, quota errors), the template silently falls back to the server-side defaults — no JS errors, no broken page. Visitors on shared/embedded browsers (kiosks, smart TVs) just get the defaults, which is the safe choice.

Templates: lightbox controls

Lightbox controls

The lightbox overlay has 4 buttons + 2 text labels:

Element	Position	Behavior
<code>.lb-close (×)</code>	Top-right (inside <code>.lb-controls</code> pill)	Closes the lightbox; Esc key
<code>.lb-open (□)</code>	Top-right (inside <code>.lb-controls</code> pill, left of close)	Opens the current image/video in a new tab
<code>.lb-prev (◀)</code>	Full-height hit area on the left side (120px wide)	Previous image; Left-arrow key
<code>.lb-next (▶)</code>	Full-height hit area on the right side (120px wide)	Next image; Right-arrow key
<code>.lb-counter</code>	Bottom-left (or bottom-center on mobile)	“N / total” text
<code>.lb-caption</code>	Bottom-center	The current file’s name

The `.lb-controls` rounded pill (Phase 48): the open (□) and close (×) buttons are wrapped in a single flex container `.lb-controls` so they appear as a single rounded pill at the top-right of the lightbox. The container has: - Position: `top: 1rem, right: 1.5rem` - Background: `rgba(255, 255, 255, 0.92)` (light pill on the dark lightbox) - Border: `2px solid #000` - Border-radius: `10px` (the “rounded box”) - Padding: `4px` around the buttons - Display: `flex` (lays out open + close side-by-side, perfectly aligned vertically and horizontally)

The two buttons inside have: - position: `static` (flex lays them out, not absolute) - Transparent background (the container provides it) - No individual border - `28×28 px` - Dark icon color (`#1a1a26`) - border-radius: `6px` on each button (rounded corners within the pill) - Hover: subtle dark background tint (`rgba(0, 0, 0, 0.08)`)

Why a pill instead of separate buttons: a single rounded container gives the two related actions (close + open) a visual unity — they’re both “exit the lightbox” actions (close or view-on-its-own), and grouping them makes that clearer.

The prev/next buttons are NOT in the pill — they’re at the left/right edges of the lightbox. These are navigation actions (different from exit actions), so they get their own positioning.

Prev/next hit areas (Phase 65): the prev/next buttons are full-window-height × 120px wide hit areas positioned at `left: 0 / right: 0`. The arrow icon is flex-centered inside the hit area. At rest, the hit area is transparent (no visible button) — only a hover reveals the target.

- Hover background — theme-aware:
 - Dark mode (default; the page bg is dark): the hover bg is `rgba(255, 255, 255, 0.08)` — a subtle whiter tint over the dark lightbox. The user sees a soft “highlight” where they’re pointing.
 - Light mode (page bg is light): the hover bg is `rgba(0, 0, 0, 0.06)` — a subtle darker tint. The lightbox itself is still theme-independent (always dark), but the hover tint adapts so it works for visitors on light pages.

Why a full-height hit area: - On touch devices (mobile, tablets), the user doesn’t have a precise cursor — a small button in the middle of the screen is hard to hit. A full-height strip on each side gives a large, easy target. - On desktop, the same hit area means the user can click anywhere in the left or right strip — no need to aim.

Why 120px wide: enough to be a comfortable target on touch screens (~7mm at typical DPI), small enough to leave the center ~60% of the screen clear for the image/video content.

Why transparent at rest: the lightbox is about the content. A visible button on each side would distract from the image. The hover reveals the hit area, so the user sees it when they’re navigating, not when they’re viewing.

JS — `stopPropagation` (Phase 65 fix): the new hit areas sit on top of the media (z-index: 1). The media element has its own click handler that advances to the next image (for `<img>`). Without `stopPropagation` on the prev/next click handlers, a single click in the prev/next area would advance TWICE — once from the button, once from the bubbling media click. The fix: both prev/next handlers call `e.stopPropagation()` before calling `show(idx±1)`.

Why an alpha-blended hover (`rgba`) instead of a solid color: the lightbox shows whatever media is loaded. A solid bg color would clash with some images (e.g., a green hover on a photo of a red rose). An alpha-blended fill mixes with whatever’s behind, giving a consistent “tint” effect that works on any media.

Why a dark border: like the tile `.open-btn`, the pill has a black 2px border so it stands out clearly against the dark lightbox background. The light pill on a dark bg with a dark border is a strong visual “this is a control” signal.

Keyboard navigation: - Escape `[]` close - Left arrow `[]` previous image - Right arrow `[]` next image - (Clicking on the image itself also goes to next, like carousel UIs)

The `data-theme` attribute on `<html>` is NOT read by the lightbox CSS — the lightbox is intentionally theme-independent (dark always, with light controls). This is the same design choice as the lightbox bg and counter/caption: focus on the content, regardless of page theme.

Video poster (Phase 63): when a video tile has a generated thumbnail (Phase 62’s `ffmpeg` pipeline), the lightbox video element sets its HTML5 poster attribute to the thumb URL (extracted from the same `<img class="thumb-img">` element on the tile). The browser shows the poster image immediately when the video opens in the lightbox — the user sees the video’s first frame as a still image, then on click the video swaps to playback. This is the same mechanism YouTube uses to show a thumbnail before a video plays.

If `no_video_thumbs` is set OR `ffmpeg` is missing, the tile has no `<img class="thumb-img">` so the JS can’t find a poster URL — the poster attribute is simply not set, and the browser shows its default (black frame, or the first decoded frame if `preload="metadata"` is enabled).

The poster URL points at the same cached WebP that’s used for the tile thumbnail, so no extra server work or storage is required.

Mobile: video play button fix (Phase 61)

On mobile devices, clicking the play button on a video in the lightbox used to advance to the next media file instead of starting playback. Root cause: the media click handler always ran `show(idx + 1)` regardless of media type. On mobile, tapping the video's native play button fires a click event on the `<video>` element, and the handler advanced to the next file BEFORE the video could play. Fix (Phase 61): check `currentEl.tagName === 'VIDEO'` in the handler and bail out so the browser's native click handling (play/pause) takes over.

For images, click-to-advance still works (unchanged). For videos, click is no longer hijacked — the user can tap the play button on mobile to start playback, or click elsewhere on the video (e.g. the time bar / volume) for the native controls to handle. Navigation is via the prev/next buttons or arrow keys.

Open-in-new-tab button (Phase 47+52)

The tile-level open-in-new-tab button (the small `⌵` in the top-right of each tile) is documented separately in its own section above. Recent refinements:

- Phase 47 — added the dark border (`border: 2px solid #000`) so the button stands out more. The light translucent bg (`rgba(255, 255, 255, 0.85)`) stays light over both light and dark page bgs, so a dark border is always visible.
- Phase 52 — arrow color fixed at `#111111` (was `var(--fg)`). The user explicitly wanted `--fg` UNCHANGED (still `#111111` light / `#e5e5e5` dark) and used by other elements. The open-btn arrow stays dark in both modes because the button's bg is always light.

Templates: header, pagination, footer, and layout features

Header meta line format

The page header (below the page title) shows a meta line summarizing the directory's contents. The format (Phase 44/45/55) is:

1	34 images · 8 videos · 2 other files // (8.3 MB total) // 4 directories · 50 per page ·					
2	Page 1 of 2					
3	count	count	count	total size	count	page size page
4	indicator			(ALL files:		
5	(when			images + videos		
6	TotalPages			+ other files)		
	> 1)					

Note the format after the `//` size segment: the directory count uses a single space separator (not `·`), then the rest of the items use `·`. This was changed in Phase 55 — the size is visually marked as “special” by the `//` framing, and the regular meta items after it use a single character (`·` or space) consistently.

Meta items, in order:

1. Image count — `N` images. Always shown.
2. Video count — `· N` videos. Shown only when `N > 0`.
3. Other files count — `· N` other files. Shown only when `N > 0`.

4. Total size — // (X.X KB total) //. Wrapped in // separators (visually distinct from the . separator used for the file counts). Represents the SUM of Size for all files in the directory: images + videos + other files. Excludes subdirectories. The literal word total follows the size inside the parens, making the meaning clear. Pre-formatted via the humanSize() helper (B / KB / MB / GB). After the closing //, the next meta item is separated by a single space (not .).
5. Directory count — N directories (single space separator after //, then the count). Shown when the user is in a subdir (N = subdirs of the current dir) or when there are subdirs in the root listing.
6. Per page size — . 50 per page. Always shown.
7. Page indicator — . Page X of Y. Shown only when TotalPages > 1 (multi-page gallery).

Why the size uses // separators instead of .: the // visually distinguishes the size from the other meta items (which all use .). The size is conceptually different — it's a quantity (bytes), not a count. The // is a “this is special” marker.

Implementation: the meta line is rendered by the template at the top of the <style> block's wrapping <header>. The size segment uses three separate elements (one for each //, one for the parens) so the browser's flex gap: 0.5rem adds visual spacing between them.

Open-in-new-tab button

The template has a small  button on each image/video tile (top-right corner of the card) that opens the file's URL in a new browser tab. Added in Phase 47; refined in Phase 52.

Visual: - Position: top: 6px, right: 6px inside the .card - Size: 28×28 px - Background: rgba(255, 255, 255, 0.85) — light translucent - Dark border: border: 2px solid #000 (Phase 47) so the button stands out more - Arrow color: #111111 (Phase 52) — a fixed dark color, NOT the theme-aware --fg token - The character itself is north east arrow (U+2197 NORTH EAST ARROW, displayed as the arrow glyph in the bundled template) - Default opacity: 0.5 (subtle). On hover/focus, opacity goes to 1.0 and the button scales up slightly.

Why the arrow color is fixed #111111 (not var(--fg)): the button has a light translucent background (rgba(255,255,255,0.85)) that stays light over ANY page background (light or dark). A dark arrow on a light button background is always visible — no need to adapt to the page's theme. The --fg token stays at its current values (#111111 light / #e5e5e5 dark) and continues to be used by other elements (hl, body, meta, sort buttons, etc.) — only .open-btn is excluded from the theme-aware color rule.

Behavior: - Click or Enter on the button  window.open(href, '_blank', 'noopener,noreferrer') opens the file in a new tab - The noopener,noreferrer flags are a security best practice (prevents the new tab from accessing window.opener) - The click event is stopPropagation'd so it doesn't also trigger the parent's “open in lightbox” handler

Pagination

The pagination nav is rendered in two places (Phase 54):

1. Top — after the sort-bar (Name/Type/Modified/Size buttons), before the DIRECTORIES section
2. Bottom — after the IMAGES grid (existing position)

Both pagination navs are identical — same buttons (Prev, page numbers, Next), same styling, same conditional (only shown when TotalPages > 1). Only the position differs.

Why mirror:

- Visitors on a long page don't have to scroll back to the top to switch pages — they can click `Next` at the top.
- The pagination at the top also serves as a “you are here” indicator when arriving on a non-first page (so the user knows they're not on page 1).
- Symmetry: the sort-bar is at the top, the pagination at the bottom; having pagination at the top mirrors the bottom.

The same `{{if gt .TotalPages 1}}` conditional guards both instances, so single-page galleries don't show any pagination at all.

Pagination item format: uses the same Google-style pattern introduced in Phase 29 — `First` (when far from start), `Prev`, page numbers with ellipsis in the middle, `Next`, `Last` (when far from end). The `aria-pressed` is set on the current page number for accessibility.

Footer

At the bottom of every gallery page, below the IMAGES grid (and below the bottom pagination if multi-page), a small footer credits the underlying technologies (Phases 56/58/59):

```
1 

---


2
3   proudly served by caddy + synapticloop // media gallery
```

- “caddy” — links to <https://caddyserver.com> (the web server)
- “synapticloop // media gallery” — links to https://github.com/synapticloop/caddy_media_gallery (this plugin's repo)

Both links have `rel="noopener" target="_blank"` (security best practice for new-tab links).

Styling: centered, small text (0.8rem), muted color (`var(--fg-muted)`), subtle padding. No border-top (removed in Phase 58 — the muted color provides enough visual distinction).

Why these credits: - “caddy” — Caddy is the web server. Without it, this plugin wouldn't exist; credit the underlying tech. - “synapticloop // media gallery” — links to this plugin's repo (not just the synapticloop org page) so visitors can read the source, file issues, or fork the project.

The footer text is operator-visible but not configurable — if you want to change it, edit the template (`<footer class="site-footer">` in the bundled template). Operators who fork the project can brand the footer however they like.

Section heading: “MEDIA” (not “IMAGES”)

The gallery renders both images AND videos (since Phase 25 when the lightbox got video support). To reflect the broader scope, the IMAGES section heading was renamed to MEDIA in Phase 60.

In the bundled template:

```
1 <h2 class="section-heading">Media</h2>
```

If you fork the template and want to revert to “IMAGES” (e.g., you have a strictly-photo gallery with no videos), change this line. The CSS (`.section-heading`) is unchanged — only the visible text.

The other section headings remain “Directories” and “Other files” (unaffected by Phase 60).

Templates: building the PDF locally

Building the PDF locally

The PDF book (caddy-media-gallery-book.pdf) is built from the markdown files in docs/ via pandoc + xelatex. The full command is wrapped in a script at the project root: build-docs.sh.

To rebuild the PDF locally:

```
1 ./build-docs.sh
```

The script: - Works from any directory (uses BASH_SOURCE to find its own location and the project root) - Verifies prerequisites (pandoc, xelatex) before running - Verifies all source files exist before running - Runs pandoc with the right flags (--pdf-engine=xelatex, --include-in-header=preamble.tex, --listings) - Cleans up intermediate files (*.aux, *.log, etc.) after - Prints summary info (output path, file size, page count)

Prerequisites: - pandoc - xelatex (usually in the texlive-xetex package) - Install on Debian/Ubuntu: sudo apt-get install pandoc texlive-xetex texlive-fonts-recommended - Install on macOS: brew install pandoc mactex

Why a script instead of running pandoc directly: the pandoc command has several flags (--include-in-header=preamble.tex references the font config; --listings is needed for code block formatting; the source file order matters). The script encapsulates all of this so you don't have to remember the incantation.

The font config (preamble.tex + docs/fonts/) references absolute paths for fontspec's Path = option. If you move this project to another machine, update the Path = line in preamble.tex to match the new location.

Sort & Pagination

The gallery's image grid respects a small set of URL query parameters. All are optional. They're a stable URL API — bookmarking ?sort=size&order=desc&page=3 works and gives the same view on every visit.

Query parameters

Param	Values	Default	What it does
?sort=	name / type / size / mtime	mtime	What field to sort images by. The URL also accepts ?sort=date (treated as ?sort=mtime) for back-compat. See "Aliases" below.
?order=	asc / desc	desc	Sort direction. desc for mtime is newest-first; desc for name/type/size is Z-A / largest-first.

Param	Values	Default	What it does
?page=	integer >= 1	1	Which page of results to show. 1-based. Out-of-range or non-numeric values fall back to 1.
?page_size=	any value in the operator-configured page_sizes list	first item	Per-page size (driven by the dropdown). Changing this resets the visitor to page 1 (the current ?page= is dropped from the form's hidden inputs). Unknown values fall back to the first item.
?q=	free text (URL-encoded)	(none)	Server-side filename search. The match rule (word or substring) is operator-configured via search_match.
?type=	comma-separated extensions (e.g. jpg,png)	(none)	Server-side type filter. The form-submission version uses repeated ?ext=jpg&ext=png (both work).

The URL parameters are preserved across changes: clicking “Name” on ?sort=mtime&order=desc&page=2 becomes ?sort=name&order=asc&page=2 (the toggle flips the order for the new field; the page stays the same). Same for the type filter, the search box, and the page-size dropdown — all preserve the other params via hidden inputs in their forms.

The directory strip and other-files strip have their own sort behaviour:

- Directories: always rendered, case-insensitive alphabetical. The Directories table has its own click-to-sort headers (name, # items, # sub-dirs, size, modified). Sort state

Size sort units (2026-07-03): The Size column shows human-readable sizes like “58.2 MB” or “1.05 GB”. The JS sort code parses these to bytes via a `parseSizeToBytes()` helper before comparing, so KB/MB/GB/TB boundaries sort correctly (e.g. 1.05 GB > 791.3 MB > 360.6 MB, not the wrong order that a naive `parseFloat` would give).

Last Modified sort fix (2026-07-03): previously all directory rows had `data-date=""` because the `ModTime` field was only populated for “other” files (not for directories, images, or videos). Fixed in `buildFileView()` by setting `v.ModTime = f.ModTime` for ALL file kinds before the switch statement. Now sorts correctly by modification time. persists in `localStorage` (per table) and in the URL (`?dirs_sort=...&dirs_order=...`). See [docs/03d-feature-localstorage.md](#) for the full key reference. - Other files: scanner order (newest-first by mtime). The Other Files table has its own click-to-sort headers (name, type, size, modified). Same persistence as Directories (see the `localStorage` reference).

The main image sort does NOT affect the dirs/other-files sorts. This is intentional: the dirs strip is a navigation aid and should be stable, not reshuffle when the user changes the image sort.

Sort indicator in the header

The header shows the current sort + a reset link:

- On the default sort (`mtime desc`), the indicator is a plain `<span>` — clicking it would just reload the same page, so it's not a link. It reads `Sort: Modified  $\downarrow$` .
- On any other sort, the indicator is a link to `?` (no query params, which is the default). It reads e.g. `Sort: Size  $\uparrow$` .

Sort buttons

The sort bar at the top of the header has four buttons: Name, Type, Modified, Size. Each shows a $\uparrow$ or $\downarrow$ arrow if it's the currently active sort. Clicking the active button toggles the order; clicking an inactive button switches to that field at the direction that's the “natural” opposite of the current order (so you don't get the same direction twice in a row).

The button labels are produced by the `sortLabel` template function (see the Templates index page, which links to the per-topic pages) doc for the full map). To add a new sort field:

1. Add a case to `sortLabel` in `render.go`
2. Add a case to the `sortFiles` function (the sort implementation)
3. Add a button to the `gallery.tmpl` template

It's a code change in three places; the template is just the visible button.

Pagination

The per-page size is configured via the `page_sizes` Caddyfile directive (default: `60 30 120 all`). The FIRST item in the list is the default per-page count (used when `?page_size=` isn't in the URL). The dropdown is rendered in the meta line of the gallery HTML so the visitor can switch the per-page size live.

Page-size change resets to page 1. When the visitor changes the per-page size via the dropdown, the form omits the `?page=` hidden input. The next page load is page 1 — the visitor doesn't end up on a non-existent page (e.g. changing from 60-per-page to 120-per-page on page 2 of a 100-image dir would otherwise land on an empty page 2).

The “Showing 1-60” range in the media header shows the visitor the slice of items they're seeing — e.g. for a 200-image gallery, page 2 shows “Media (200 - Showing 61-120)”.

Pagination links in the live template: - The “First” and “Prev” pagination buttons (rendered by the template with double- and single-left-arrow icons, when `{{.HasPrev}}` is true) - The “Next” and “Last” pagination buttons (single- and double-right-arrow icons, rendered when `{{.HasNext}}` is true) - Current page indicator in the middle

The page numbers in the URL are 1-based, not 0-based. `?page=0` falls back to `?page=1`.

Search

The search box in the header does two things:

- As the visitor types (client-side filter): items that don't match get the `.filtered-out` class. Visibility collapse + opacity 0 fade hides them. The URL doesn't change.

- “Search All” button (server-side filter): submits the form with `?q=foo`. The page re-loads with the matched files only. The form preserves the other URL params.

The match rule is the same on both sides: `word` (default “substring” without config) or `word` (when the operator sets `search_match word`). The visitor doesn’t need to know — the match just works.

Type filter

The “File Type Filter” dropdown has checkboxes for Images / Videos / Other, with the count of each type next to the label. Checking “Images” + “Videos” (unchecking “Other”) filters out non-media files. The “Reset” pill resets the filter.

The (none) entry — the Other dropdown includes a special (none) option for files that don’t have an extension (e.g. `Makefile`, `welcome`). When checked, ONLY files without an extension are shown — no `.md`, no `.txt`, no other recognized extension. The form value is the sentinel `.` (a literal dot) which can’t be a real file extension; `parseTypeFilter` translates it to the empty-string filter key, and `applyTypeFilter` checks `filter[""]` to apply the strict filter. Multi-select works as expected: checking (none) + `.md` shows files matching either condition.

The filter is applied both server-side (when the form is submitted) and via the URL (`?type=jpg&type=png`). Directories are never filtered (the user can always navigate).

Examples

```
1 /images/                                # default: mtime desc, page 1
2 /images/?sort=name&order=asc            # alphabetical
3 /images/?sort=size&order=desc           # largest first
4 /images/?page=3                         # third page of default sort
5 /images/?sort=type&order=asc&page=2
6 /images/?page_size=120                  # change per-page to 120
7 /images/?q=cat                          # server-side search for "cat"
8 /images/?type=jpg,png                   # only JPG and PNG
```

Aliases

`?sort=date` is an alias for `?sort=mtime` — both sort by the file’s modification time. The two values produce identical results:

```
1 /images/?sort=date&order=desc           # = /images/?sort=mtime&order=desc
2 /images/?sort=date&order=asc            # = /images/?sort=mtime&order=asc
```

Why have the alias? The two names describe the same thing from different angles — “mtime” is the technical field name (modification time, from the filesystem), “date” is the user-friendly name (the date the file was last changed). The “Modified” sort button in the UI emits `?sort=mtime` (the technical name), but the URL API accepts both for back-compat with bookmarks and external links that might use either name.

Implementation: `parseSort` in `render.go` accepts “date” in its switch statement and treats it identically to “mtime”:

```
1 switch field {
2 case "name", "type", "date", "mtime", "size":
3     // all valid; "date" is an alias for "mtime"
```

```
4 default:
5     field = "mtime" // unknown fields fall back to the default
6 }
```

In `sortFiles`, both `"mtime"`, `"date"`, and `""` (the empty default) share the same sort branch — sort by `ModTime` per the `order` param. So functionally there's no distinction between them.

When to use which: if you're writing a new integration or bookmark, prefer `?sort=mtime` (it's the canonical name and what the UI button emits). `?sort=date` is fine too — just pick one and stick with it for consistency.

Stability guarantees

- The URL parameter API is stable. Bookmarking works. The four sort fields, two orders, the search query, the type filter, and the per-page size are part of the public contract; renaming or removing one is a breaking change.
- Adding a new sort field is non-breaking (just adds a new option).
- The default sort may change in a future major version, but the `?sort=mtime` URL will always be honored.
- Page size may change between versions, but URLs with `?page=N` are stable relative to the *current* page size — if the page size changes from 60 to 100, page 3 of the old URL might now be page 2 of the new layout, but no image is lost or duplicated (it's the same image set, just packed differently).
- Search match mode (`?q=...`) is stable; the operator can change the match rule (word/substring) at any time and old bookmarks still work the same way.

Font credits

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Project page: <https://www.jetbrains.com/lp/mono/>

Bundled location

The TTF files for both fonts are bundled in this repository at `docs/fonts/`:

File	Family	Style
LibreBaskerville-VF.ttf	Libre Baskerville	Regular (variable weight 400-700)
LibreBaskerville-Italic-VF.ttf	Libre Baskerville	Italic (variable weight 400-700)
JetBrainsMono-VF.ttf	JetBrains Mono	Regular (variable weight 100-800)
JetBrainsMono-Italic-VF.ttf	JetBrains Mono	Italic (variable weight 100-800)

The VF suffix indicates the variable-font axis. Both fonts include the full weight range in a single file, so a single fontset declaration in the LaTeX preamble handles all weights without needing separate files for bold/italic.

How the fonts are loaded

The fonts are loaded by `docs/preamble.tex` (the LaTeX preamble included in every page of this PDF) via `\setmainfont` (for Libre Baskerville) and `\setmonofont` (for JetBrains Mono), both with a relative `Path = ./fonts/` so the preamble is portable across users (no hardcoded `/home/<user>/...` paths in the committed file).

If you want to replace the fonts with alternatives:

1. Drop your TTF/OTF files in `docs/fonts/` (using the same filenames so the preamble doesn't need to change), OR
2. Edit `docs/preamble.tex` to point at your font files and re-run `./build-docs.sh`.

The bundled fonts are licensed under SIL OFL 1.1 (free for commercial use, modification, and redistribution). Any substitution is your responsibility — ensure the replacement font's license is compatible with the project.

05. End Plate

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